

**Big picture goal:** develop a general pipeline in which an arbitrary protein can be segmented into all of its possible epitopes, and then as many of these epitopes as possible faithfully represented in the form of scaffolded designs that can be used to detect antibodies against the parent protein

**Immediate goal:** design a new PepSeq library ("DP4") consisting of ~6,000-12,000 103mers (depending on how many we can justify) that helps provide initial answers to 3 key questions:

Question 1: What is a minimal epitope? How many islands / AAs are generally needed for binding?

- (focused on the 57 mAb epitopes)
- scaffold the islands individually (which may increase success rate)
- scaffold the islands together but knock out individual islands with Alanines (per DP3)
- set different radii to define more v less inclusive AA sets within epitopes?

Question 2: What is required to successfully scaffold a minimal epitope? Given designs targeting a focal epitope, what features (epitope v scaffold RMSD, PAE, clash etc) predict binding?

- (focused on the 57 mAb epitopes)
- generate a large number of designs across epitopes and then take a sampling of these designs (from across the RMSD/PAE/clash space) into PepSeq and see what binds.
  - You have existing RFD3-generated designs on these epitopes, but if compute capacity allows, consider expanding the sampling (potentially at all 3 levels: (i) contig construction, (ii) RFD, (iii) MPNN), especially for antibodies that were under-sampled from Lawson's set.

Question 3: Can the approach be extended to the polyclonal setting? Given an entire protein, can we develop an implementation of steps #1 and #2 to successfully identify and scaffold 100s of possible epitopes across the protein, and thereby detect reactivity in polyclonal serum samples where only the protein-level (not epitope-level) reactivity is known?

- given a protein, take all possible tiles (e.g. overlapping 12mers?) across the linear sequence of the protein (or at least the solvent accessible regions) and scaffold them individually
- as an alternative, apply "spongy ball" algo to identify (multi-island) epitopes that are on the surface and spatially clustered
- apply these approaches (and/or any others) to the following target proteins (for which we have positive and negative samples in house):
  - 1D2K – Cocci CF antigen 951, 317
    - ~200 12mer tiles @step=2
  - 6M0J (chain B) – SARS-CoV-2 RBD 450, 150
    - ~110 12mer tiles @step=2
  - 4WAT – PfRH5 antigen 780, 260
    - ~200 12mer tiles @step=2
- note that for this question, we will need to generalize the clash predictor, since no antibody sequence/structure will be available

DP4 working outline:

| # | Project    | Category                   | Related to question # | Description                                                                                                                                                   | #new epitopes to scaffold | #DP4 peptides |
|---|------------|----------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|
| 1 | Episcaf    | “individual island”        | 1                     | 46 dual-island mAb epitopes x 2 individual islands X 5 “best” designs each                                                                                    | 92                        | 460           |
| 3 | Episcaf    | “tiled 30mers”             | 1                     | tile:<br>* 57 parent epitope antigens, and<br>* 3 scaffolded 12mer antigens, as 30mers, step=6 (no scaffolding, just linear tiling – as a point of reference) | 0                         | 2,085         |
| 4 | Episcaf    | “best scaffolded”          | 2                     | "best" 20 of each mAb x 57 mAbs                                                                                                                               | ≥0                        | 1,140         |
| 5 | Episcaf    | “titration of parameters”  | 2                     | 100-peptide 4-parameter "titration" x 57 mAbs                                                                                                                 | ≥0                        | 5,700         |
| 6 | Episcaf    | “scaffolded 12mer tiles”   | 3                     | tiled 12mers, step=2 for 3 proteins; 500 epitopes X 5 “best” designs each                                                                                     | 500                       | 2,181         |
| 7 | Episcaf    | CIDR domain (side project) | NA                    | scaffold the epitope from this structure:<br><a href="https://www.rcsb.org/3d-view/8VDL/1">https://www.rcsb.org/3d-view/8VDL/1</a>                            | 1                         | 10            |
| 8 | CF-GENESIS | New PfEMP1 minibinders     | NA                    | John will collect these from LX; currently have >15,000 passing filters, so will want to narrow it down                                                       |                           | 5000          |

Totals:

~650

~12,000

Leftover:

|   |                         |   |                                                                                                  |    |     |
|---|-------------------------|---|--------------------------------------------------------------------------------------------------|----|-----|
| 2 | <b>“reduced radius”</b> | 1 | 57 parent pdbs but now with epitopes redefined at a reduced radius (3Å?) X 5 “best” designs each | 57 | 285 |
|---|-------------------------|---|--------------------------------------------------------------------------------------------------|----|-----|

Note: in order to define “best”, we will need a single composite score  $f(\text{RMSDs}, \text{PAE}, \text{clash\#})$ , for example:

```
compositeLoss=  
pmax(0, (filteredFeatures["epitope_chunk_rmsd_vs_mpnn"]-1))+  
pmax(0, (filteredFeatures["overall_rmsd"]-2))+  
0.5*pmax(0, (filteredFeatures["mean_pae"]-5))+  
0.5*data["clashResidueCount"]
```